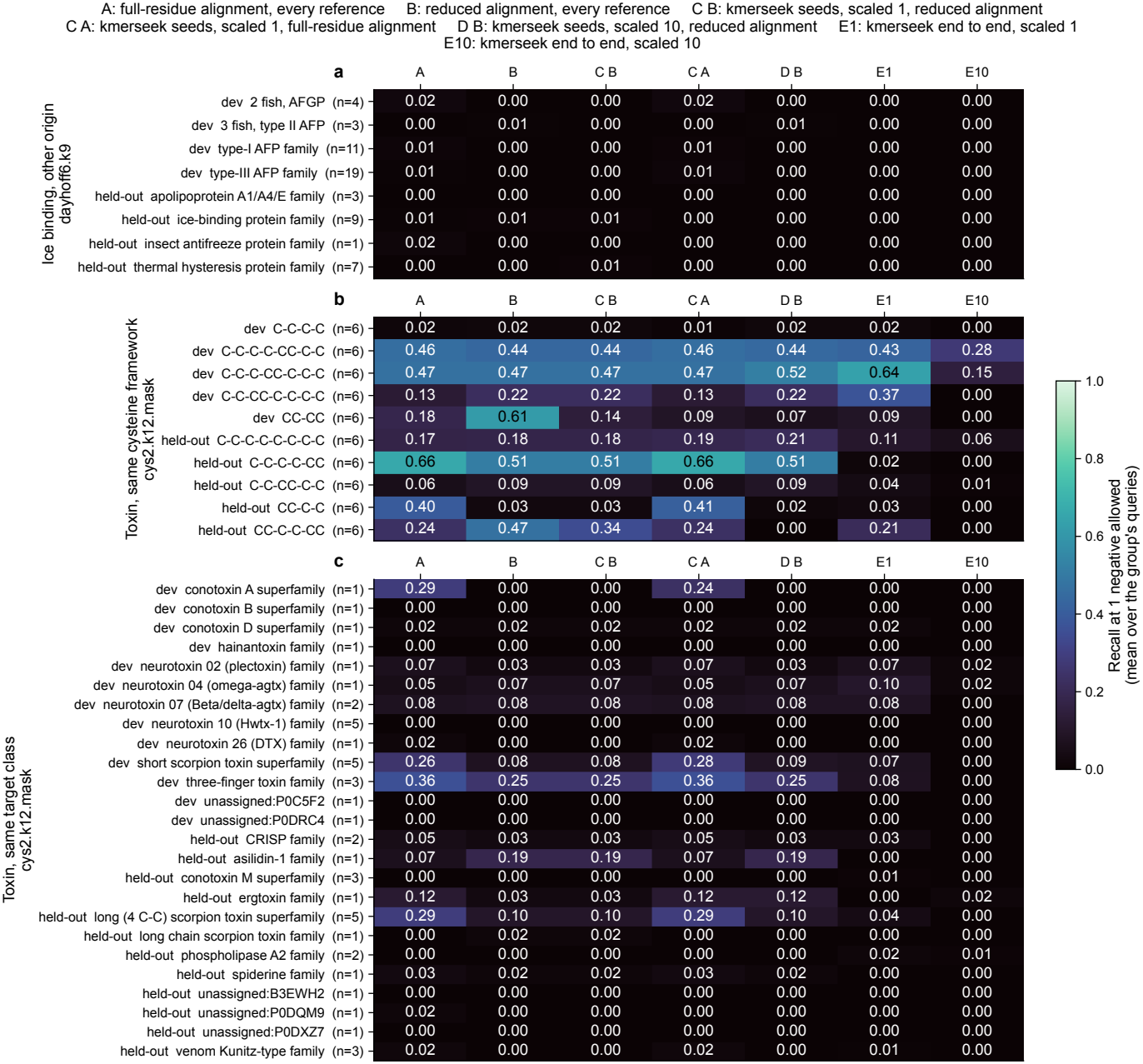

A: full-residue alignment, every reference
B: reduced alignment, every reference
C B: kmerseek seeds, scaled 1, reduced alignment
C A: kmerseek seeds, scaled 1, full-residue alignment
D B: kmerseek seeds, scaled 10, reduced alignment
E1: kmerseek end to end, scaled 1
E10: kmerseek end to end, scaled 10
a
A B C B C A D B E1 E10
Ice binding, other origin dayhoff6.k9
dev 2 fish, AFGP (n=4) 0.02 0.00 0.00 0.02 0.00 0.00 0.00
dev 3 fish, type II AFP (n=3) 0.00 0.01 0.00 0.00 0.01 0.00 0.00
dev type-I AFP family (n=11) 0.01 0.00 0.00 0.01 0.00 0.00 0.00
dev type-III AFP family (n=19) 0.01 0.00 0.00 0.01 0.00 0.00 0.00
held-out apolipoprotein A1/A4/E family (n=3) 0.00 0.00 0.00 0.00 0.00 0.00 0.00
held-out ice-binding protein family (n=9) 0.01 0.01 0.01 0.00 0.00 0.00 0.00
held-out insect antifreeze protein family (n=1) 0.02 0.00 0.00 0.00 0.00 0.00 0.00
held-out thermal hysteresis protein family (n=7) 0.00 0.00 0.01 0.00 0.00 0.00 0.00
b
A B C B C A D B E1 E10
Toxin, same cysteine framework cys2.k12.mask
dev C-C-C-C (n=6) 0.02 0.02 0.02 0.01 0.02 0.02 0.00
dev C-C-C-C-CC-C-C (n=6) 0.46 0.44 0.44 0.46 0.44 0.43 0.28
dev C-C-C-CC-C-C-C (n=6) 0.47 0.47 0.47 0.47 0.52 0.64 0.15
dev C-C-CC-C-C-C-C (n=6) 0.13 0.22 0.22 0.13 0.22 0.37 0.00
dev CC-CC (n=6) 0.18 0.61 0.14 0.09 0.07 0.09 0.00
held-out C-C-C-C-C-C-C-C (n=6) 0.17 0.18 0.18 0.19 0.21 0.11 0.06
held-out C-C-C-C-CC (n=6) 0.66 0.51 0.51 0.66 0.51 0.02 0.00
held-out C-C-CC-C-C (n=6) 0.06 0.09 0.09 0.06 0.09 0.04 0.01
held-out CC-C-C (n=6) 0.40 0.03 0.03 0.41 0.02 0.03 0.00
held-out CC-C-C-CC (n=6) 0.24 0.47 0.34 0.24 0.00 0.21 0.00
c
A B C B C A D B E1 E10
Toxin, same target class cys2.k12.mask
dev conotoxin A superfamily (n=1) 0.29 0.00 0.00 0.24 0.00 0.00 0.00
dev conotoxin B superfamily (n=1) 0.00 0.00 0.00 0.00 0.00 0.00 0.00
dev conotoxin D superfamily (n=1) 0.02 0.02 0.02 0.02 0.02 0.02 0.00
dev hainantoxin family (n=1) 0.00 0.00 0.00 0.00 0.00 0.00 0.00
dev neurotoxin 02 (plectoxin) family (n=1) 0.07 0.03 0.03 0.07 0.03 0.07 0.02
dev neurotoxin 04 (omega-agtx) family (n=1) 0.05 0.07 0.07 0.05 0.07 0.10 0.02
dev neurotoxin 07 (Beta/delta-agtx) family (n=2) 0.08 0.08 0.08 0.08 0.08 0.08 0.00
dev neurotoxin 10 (Hwtx-1) family (n=5) 0.00 0.00 0.00 0.00 0.00 0.00 0.00
dev neurotoxin 26 (DTX) family (n=1) 0.02 0.00 0.00 0.02 0.00 0.00 0.00
dev short scorpion toxin superfamily (n=5) 0.26 0.08 0.08 0.28 0.09 0.07 0.00
dev three-finger toxin family (n=3) 0.36 0.25 0.25 0.36 0.25 0.08 0.00
dev unassigned:P0C5F2 (n=1) 0.00 0.00 0.00 0.00 0.00 0.00 0.00
dev unassigned:P0DRC4 (n=1) 0.00 0.00 0.00 0.00 0.00 0.00 0.00
held-out CRISP family (n=2) 0.05 0.03 0.03 0.05 0.03 0.03 0.00
held-out asilidin-1 family (n=1) 0.07 0.19 0.19 0.07 0.19 0.00 0.00
held-out conotoxin M superfamily (n=3) 0.00 0.00 0.00 0.00 0.00 0.01 0.00
held-out ergtoxin family (n=1) 0.12 0.03 0.03 0.12 0.12 0.00 0.02
held-out long (4 C-C) scorpion toxin superfamily (n=5) 0.29 0.10 0.10 0.29 0.10 0.04 0.00
held-out long chain scorpion toxin family (n=1) 0.00 0.02 0.02 0.00 0.00 0.00 0.00
held-out phospholipase A2 family (n=2) 0.00 0.00 0.00 0.00 0.00 0.02 0.01
held-out spiderine family (n=1) 0.03 0.02 0.02 0.03 0.02 0.00 0.00
held-out unassigned:B3EWH2 (n=1) 0.00 0.00 0.00 0.00 0.00 0.00 0.00
held-out unassigned:P0DQM9 (n=1) 0.02 0.00 0.00 0.00 0.00 0.00 0.00
held-out unassigned:P0DXZ7 (n=1) 0.00 0.00 0.00 0.00 0.00 0.00 0.00
held-out venom Kunitz-type family (n=3) 0.02 0.00 0.00 0.02 0.00 0.01 0.00
Recall at 1 negative allowed (mean over the group's queries)
1.0
0.8
0.6
0.4
0.2
0.0